Chapter test

4 Movement of substances across the cell membrane

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following substances can freely pass through the phospholipid bilayer of the cell membrane?

   A. water
   B. calcium ion
   C. oxygen
   D. glucose ☐

2. Which of the following is/are needed for diffusion to occur?

   (1) a concentration gradient
   (2) a differentially permeable membrane
   (3) supply of ATP

   A. (1) only
   B. (2) only
   C. (1) and (2) only
   D. (1), (2) and (3) ☐

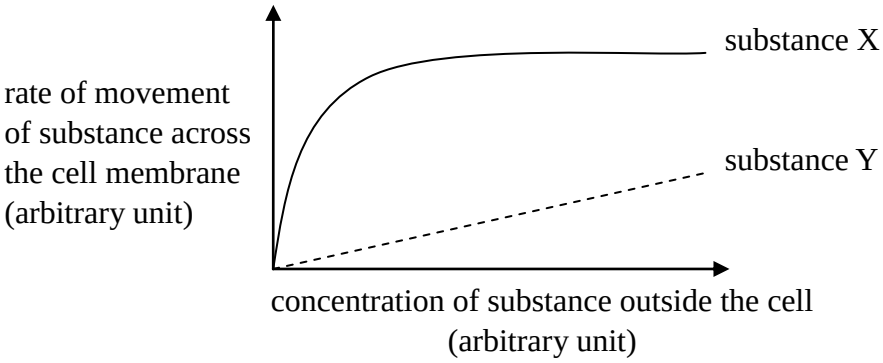
3. Which processes of moving substances across the cell membrane would result in equilibrium?

   (1) diffusion
   (2) osmosis
   (3) active transport

   A. (1) and (2) only
   B. (2) and (3) only
   C. (1) and (3) only
   D. (1), (2) and (3) ☐

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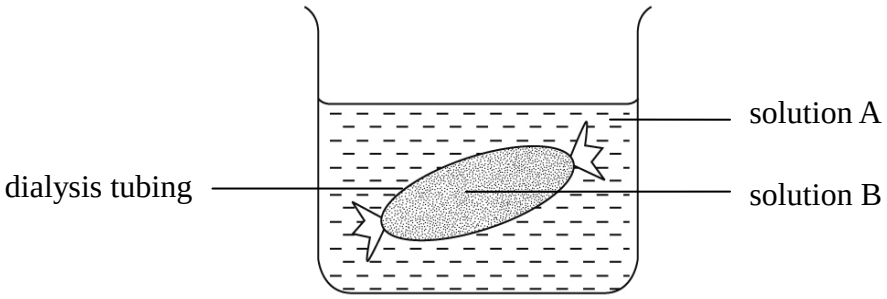
★ 4. The graph below shows the rate of movement of two substances, X and Y, across the cell membrane. X and Y move across the membrane through different mechanisms.



Which of the following can be deduced from the graph?

A. Substance X is lipid-soluble.
B. The transport of substance X involves carrier proteins.
C. The rate of transport of substance Y is not affected by temperature.
D. The rate of transport of substance Y will always be lower than that of substance X. ☐

★ 5. The diagram below shows a set-up demonstrating osmosis. The dialysis tubing used is impermeable to sucrose molecules.

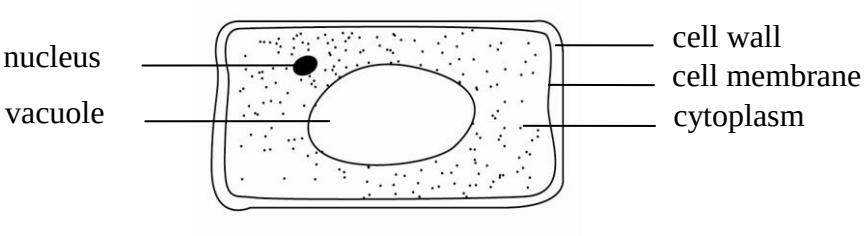


Which of the following correctly identifies solutions A and B, and the corresponding change in the volume of the dialysis tubing?

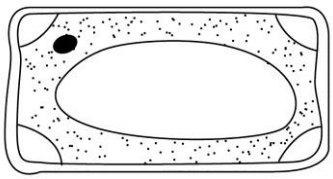
	Solution A	***Solution B***	***Change in the volume of dialysis tubing***
A.	distilled water	10% sucrose solution	decreases
B.	10% sucrose solution	distilled water	increases
C.	10% sucrose solution	20% sucrose solution	increases
D.	10% sucrose solution	10% sucrose solution	decreases ☐

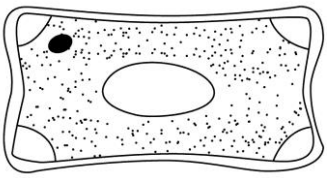
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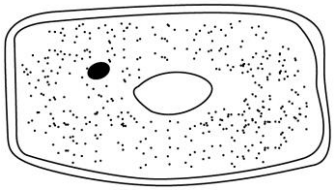
6. The diagram below shows the appearance of a plant cell immersed in an isotonic solution.

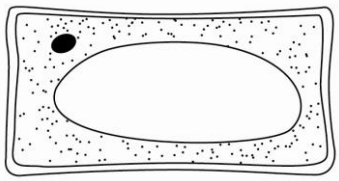


Which of the following diagrams correctly represents the appearance of the plant cell immersed in a hypertonic solution?

A. 

B. 

C. 

D. 

☐

★ 7. Which of the following is an example of passive movement across the cell membrane?

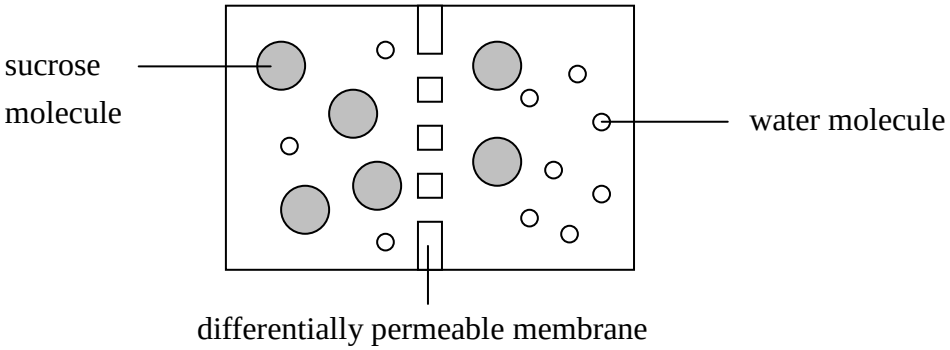
A. phagocytes engulfing a bacterium
B. *Paramecium* in a fresh water stream gaining water from the surroundings
C. some cells of the small intestine excreting mucus into the gut lumen
D. *Amoeba* feeding on algae ☐

8. Which of the following processes only occurs in living cells?

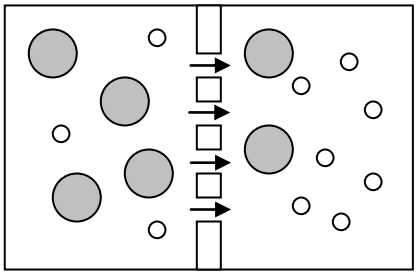
A. diffusion
B. evaporation
C. osmosis
D. phagocytosis ☐

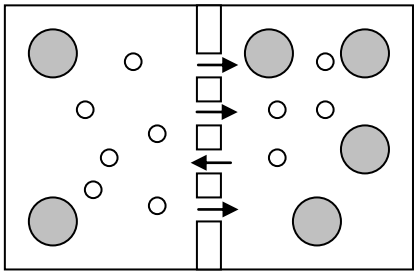
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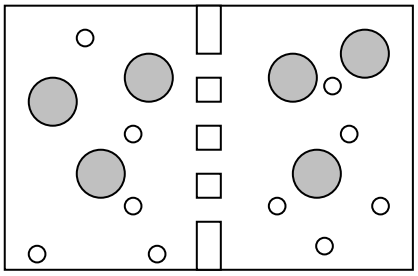
★ 9. The diagram below shows the initial distribution of sucrose and water molecules.

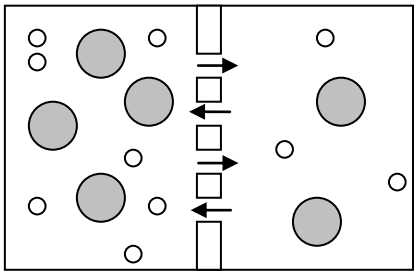


Which of the following diagrams best represents the movement of molecules at the equilibrium stage? (The movement of water molecule is indicated by the arrows)

A.


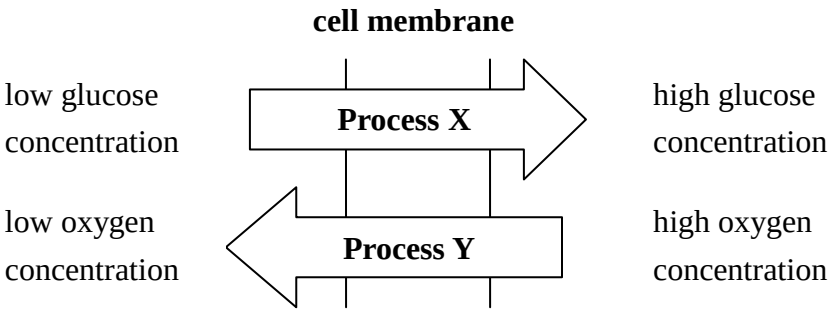
B.


C.


D.


□

★ 10. The diagram below shows two processes of transport of substances across the cell membrane.



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Identify processes X and Y.

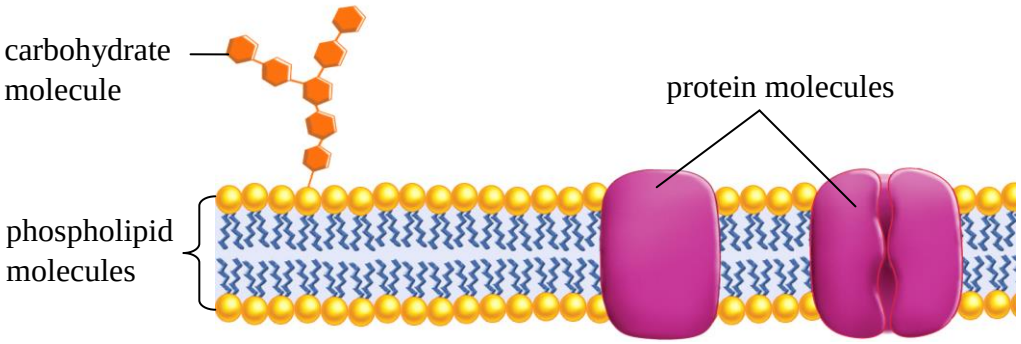
	Process X	*Process Y*
A.	diffusion	active transport
B.	active transport	diffusion
C.	diffusion	phagocytosis
D.	active transport	phagocytosis

☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	4.1	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2

B. Problem solving (12 marks)

11. Singer and Nicolson proposed the fluid mosaic model in 1972. The diagram below illustrates the model.



(a) Why is this cell membrane model named as 'fluid mosaic' model? (2 marks)

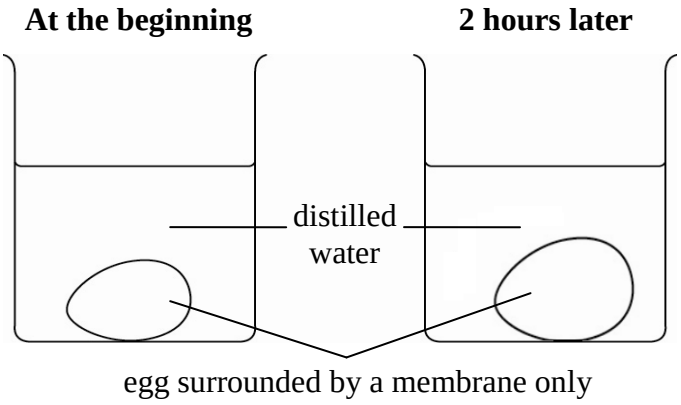
(b) Explain how the properties of phospholipid result in the formation of a bilayer. (3 marks)

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(c) Protein molecules are denatured by high temperature. Describe and explain the effect of heating on the permeability of the cell membrane. (2 marks)

(Total: 7 marks)

★ 12. A student placed an egg in a beaker of dilute hydrochloric acid for 24 hours. The egg shell was dissolved by the acid, leaving the egg surrounded by a membrane. This membrane has similar properties as the cell membrane. The egg was then immersed into distilled water. The diagram below shows the result.



(a) Describe and explain the result of the experiment. (4 marks)

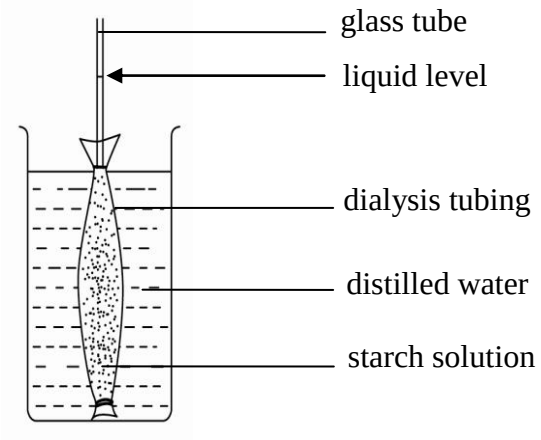
(b) Frogs carry out external fertilization. They lay eggs in fresh water, but never in sea water. The eggs are only surrounded by a membrane but are unprotected by any shell. Explain why frogs do not lay eggs in sea water. (1 mark)

(Total: 5 marks)

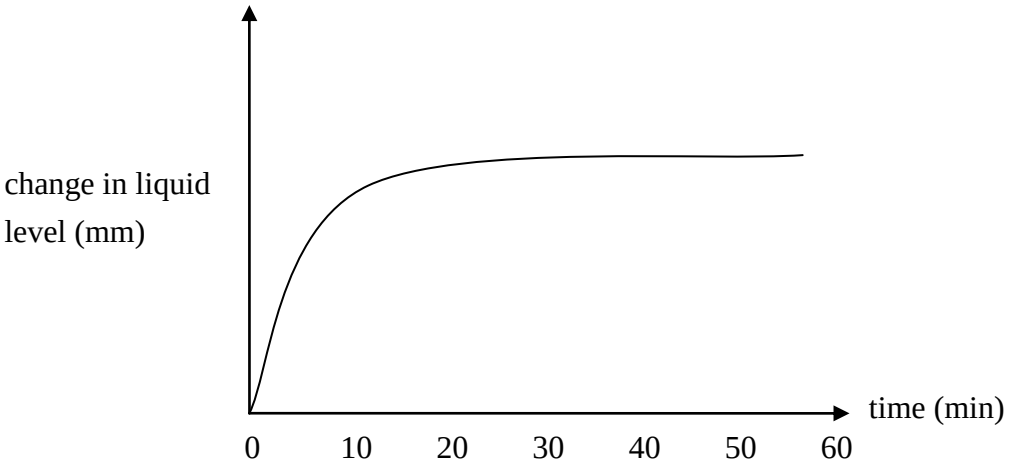
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C. Exam challenge (18 marks)

★ 13. James investigated the rate of osmosis using the set-up shown below. The dialysis tubing used in this experiment is permeable to molecules with molecular size smaller than sucrose.



The initial liquid level in the glass tube was marked as 0. The liquid level was recorded every five minutes. The graph below shows the result.



(a) (i) Describe the changes in liquid level. (3 marks)

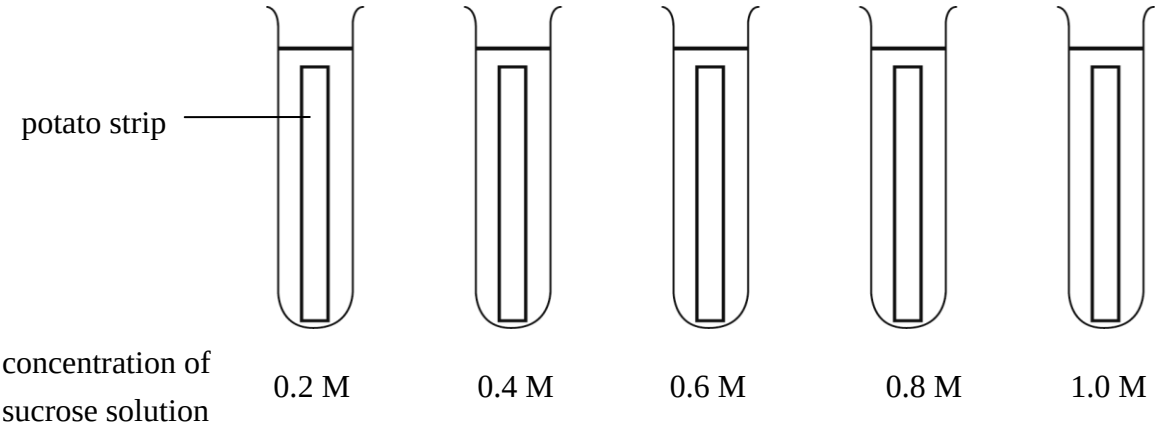
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(ii) Explain the change in the liquid level from 0 to 10 minutes. (2 marks)

(b) Predict and explain the changes in the liquid level inside the glass tube if the experiment is repeated with glucose solution instead of starch solution. (6 marks)

(Total: 11 marks)

★★ 14. Mary carried out an investigation to determine the water potential of fresh potato tissue. Potato strips with the same length were placed into test tubes with different concentrations of sucrose solutions.



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The table below shows the change of the length of the potato strips after immersion in sucrose solution for one hour.

Sucrose concentration (M)	Change in the length of the potato strip (mm)
0.2	2.0
0.4	1.2
0.6	0.6
0.8	-0.1
1.0	-1.0

(a) Briefly explain the principle of the experiment. (4 marks)

(b) State the range of concentration of sucrose solution that has the same concentration as the potato tissue. (1 mark)

(c) Mary's teacher commented that the percentage error of the length of the potato strips may be very high because the changes in length are very small.

(i) Other than measuring the length of the potato strips, suggest another way to accurately measure the change in potato strips. (1 mark)

(ii) Other than using potato strips of the same length, give ***one*** other control variables that should be kept constant so as to ensure the investigation is a fair test. (1 mark)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	C	2.	A	3.	A	4.	B	5.	C
6.	B	7.	B	8.	D	9.	D	10.	B

(1 mark each)

B. Problem solving (12 marks)

11. (a) 'Fluid' – the phospholipid molecules and some protein molecules can move laterally in the cell membrane. (1)
'Mosaic' – protein molecules are interspersed among the phospholipid molecules in a mosaic pattern. (1)

(b) The both sides of the membrane are aqueous environments. (1)
Fatty acids are hydrophobic. They orientate themselves away from the aqueous environment. (1)
Phosphate groups are hydrophilic. They orientate themselves towards the aqueous environment. (1)

(c) When carrier proteins and channel proteins are denatured by high temperature, their shapes are changed. (1)
The proteins can no longer regulate which molecules can pass through, so the membrane becomes fully permeable. (1)

(Total: 7 marks)

12. (a) The water potential of the egg content was lower than that of distilled water. (1)
There was net gain of water into the egg through the membrane (1)
by osmosis. (1)
The size of the egg therefore increased. (1)

(b) The eggs would lose water / be dehydrated by osmosis. (1)

(Total: 5 marks)

C. Exam challenge (18 marks)

13. (a) (i) The liquid level increased rapidly during the first 10 minutes. (1)
The rise of the liquid level became slower for the next 20 minutes. (1)
There was no change in the liquid level between 30 minutes and 60 minutes. (1)

(ii) The water potential gradient is the steepest at the beginning of the experiment. (1)
The water molecules enter the starch solution at the highest rate. (1)

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(b) Initially, the water potential of the glucose solution is lower than that of the distilled water. (1)
There would be a net entry of water into the tubing by osmosis, so the liquid level inside the glass tube would rise. (1)
Due to the concentration gradient, glucose molecules would diffuse out of the tubing into the beaker of water. (1)
When the water potential inside becomes the same as that of the outside, there would be no net movement of water molecules. The liquid level would stop rising. (1)
As glucose molecules continue to diffuse out down the concentration gradient, the water potential of the outside would become lower than that of the inside. (1)
There would be net movement of water molecules out of the tubing. The liquid level would fall. (1)

(Total: 11 marks)

14. (a) If the concentration of the sucrose solution is lower than that of the cell sap of the potato cells, the cells will gain water by osmosis. Therefore, the potato strips will increase in length. (1)
If the concentration of the sucrose solution is higher than that of the cell sap of the potato cells, the cells will lose water by osmosis. Therefore, the potato strips will decrease in length. (1)
If the sucrose solution is isotonic to the cell sap of the potato cells, there will be no net movement of water molecules into or out of the cells. (1)
If the potato strip shows no increase or decrease in length when immersed in a solution of known concentration, its water potential is equal that of the external solution. (1)

(b) 0.6 M > concentration of the potato tissue > 0.8 M (1)

(c) (i) measure the change in the weight of the potato strips with an electronic balance (1)

(ii) surface area of the potato strips /
volume of the sucrose solution /
temperature of the sucrose solution /
all potato strips are obtained from the same potato (Any ***one***, 1)

(Total: 7 marks)

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